# S1. Detailed structures of DL models

TABLE I
PRIMARY STRUCTURE OF THE LSTM-FCN

| Layer | Output Shape | Hyperparameters |
|---|---|---|
| **Channel 1: LSTM** | | |
| Input | $(N, 150, 88)$ | — |
| Bi-LSTM | $(N, 150, 256)$ | num_layers=3, bidir=True |
| **Channel 2: FCN** | | |
| Input | $(N, 88, 150)$ | — |
| Conv1d + BN + ReLU | $(N, 128, 150)$ | $k = 3, s = 1, p = 1$ |
| Conv1d + BN + ReLU | $(N, 256, 150)$ | $k = 3, s = 1, p = 1$ |
| Conv1d + BN + ReLU | $(N, 256, 150)$ | $k = 3, s = 1, p = 1$ |
| **Fusion & Classification** | | |
| Concatenation | $(N, 384)$ | Concat(LSTM_out, FCN_out)* |
| Linear (fc1) + ReLU | $(N, 256)$ | — |
| Linear (fc2) | $(N, 2)$ | — |
| Dropout | $(N, 2)$ | $p_{drop} = 0.5$ |
| Output | $(N, 2)$ | — |

$k$, $s$, and $p$ denote kernel size, stride, and padding, respectively. Learning_rate=0.0006.

TABLE II
PRIMARY STRUCTURE OF THE DRCA USED IN LOAD-SHEDDING CASES

| Layer | Output Shape | Hyperparameters |
|---|---|---|
| Input | $(N, 79, 150)$ | — |
| Conv1d | $(N, 128, 150)$ | $k = 3, s = 1, p = 1$ |
| ReLU | $(N, 128, 150)$ | — |
| ResBlock1D | $(N, 128, 150)$ | $k = 3, s = 1$ |
| Conv1d | $(N, 128, 75)$ | $k = 3, s = 2, p = 1$ |
| ReLU | $(N, 128, 75)$ | — |
| ResBlock1D | $(N, 128, 75)$ | $k = 3, s = 1$ |
| Conv1d | $(N, 256, 38)$ | $k = 3, s = 2, p = 1$ |
| ReLU | $(N, 256, 38)$ | — |
| ResBlock1D | $(N, 256, 38)$ | $k = 3, s = 1$ |
| Conv1d | $(N, 512, 19)$ | $k = 3, s = 2, p = 1$ |
| ReLU | $(N, 512, 19)$ | — |
| ResBlock1D | $(N, 512, 19)$ | $k = 3, s = 1$ |
| Conv1d | $(N, 512, 10)$ | $k = 3, s = 2, p = 1$ |
| ReLU | $(N, 512, 10)$ | — |
| ResBlock1D | $(N, 512, 10)$ | $k = 3, s = 1$ |
| Transposed1d | $(N, 512, 20)$ | $k = 3, s = 2, p = 1, op = 1$ |
| ReLU | $(N, 512, 20)$ | — |
| ResBlock1D | $(N, 512, 20)$ | $k = 3, s = 1$ |
| Transposed1d | $(N, 256, 40)$ | $k = 3, s = 2, p = 1, op = 1$ |
| ReLU | $(N, 256, 40)$ | — |
| ResBlock1D | $(N, 256, 40)$ | $k = 3, s = 1$ |
| Transposed1d | $(N, 128, 80)$ | $k = 3, s = 2, p = 1, op = 1$ |
| ReLU | $(N, 128, 80)$ | — |
| ResBlock1D | $(N, 128, 80)$ | $k = 3, s = 1$ |
| Transposed1d | $(N, 128, 160)$ | $k = 3, s = 2, p = 1, op = 1$ |
| ReLU | $(N, 128, 160)$ | — |
| ResBlock1D | $(N, 128, 160)$ | $k = 3, s = 1$ |
| Conv1d | $(N, 79, 160)$ | $k = 3, s = 1, p = 1$ |
| Sigmoid | $(N, 79, 160)$ | — |
| Output | $(N, 79, 150)$ | time_step=150 |

$k$, $s$, $p$, and $op$ denote kernel size, stride, padding, and output padding, respectively. Learning_rate=0.0005.

TABLE III
PRIMARY STRUCTURE OF THE DRCA USED IN GENERATOR-TRIPPING CASES

| Layer | Output Shape | Hyperparameters |
|---|---|---|
| Input | $(N, 79, 150)$ | — |
| Conv1d | $(N, 256, 150)$ | $k = 3, s = 1, p = 1$ |
| ReLU | $(N, 256, 150)$ | — |
| ResBlock1D | $(N, 256, 150)$ | $k = 3, s = 1$ |
| Conv1d | $(N, 512, 75)$ | $k = 3, s = 2, p = 1$ |
| ReLU | $(N, 512, 75)$ | — |
| ResBlock1D | $(N, 512, 75)$ | $k = 3, s = 1$ |
| Conv1d | $(N, 512, 38)$ | $k = 3, s = 2, p = 1$ |
| ReLU | $(N, 512, 38)$ | — |
| ResBlock1D | $(N, 512, 38)$ | $k = 3, s = 1$ |
| Conv1d | $(N, 1024, 19)$ | $k = 3, s = 2, p = 1$ |
| ReLU | $(N, 1024, 19)$ | — |
| ResBlock1D | $(N, 1024, 19)$ | $k = 3, s = 1$ |
| Transposed1d | $(N, 512, 38)$ | $k = 3, s = 2, p = 1, op = 1$ |
| ReLU | $(N, 512, 38)$ | — |
| ResBlock1D | $(N, 512, 38)$ | $k = 3, s = 1$ |
| Transposed1d | $(N, 512, 76)$ | $k = 3, s = 2, p = 1, op = 1$ |
| ReLU | $(N, 512, 76)$ | — |
| ResBlock1D | $(N, 512, 76)$ | $k = 3, s = 1$ |
| Transposed1d | $(N, 256, 152)$ | $k = 3, s = 2, p = 1, op = 1$ |
| ReLU | $(N, 256, 152)$ | — |
| ResBlock1D | $(N, 256, 152)$ | $k = 3, s = 1$ |
| Conv1d | $(N, 79, 152)$ | $k = 3, s = 1, p = 1$ |
| Sigmoid | $(N, 79, 152)$ | — |
| Output | $(N, 79, 150)$ | time_step=150 |

$k$, $s$, $p$, and $op$ denote kernel size, stride, padding, and output padding, respectively. Learning_rate=0.0001.

## S2. Training loss

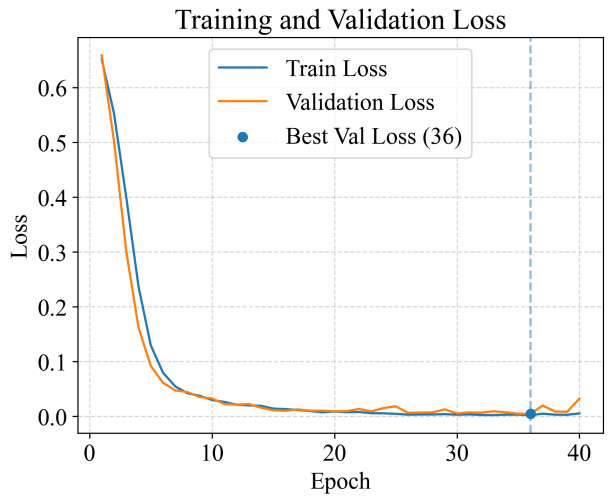


Fig. 1. LSTM-FCN training loss.

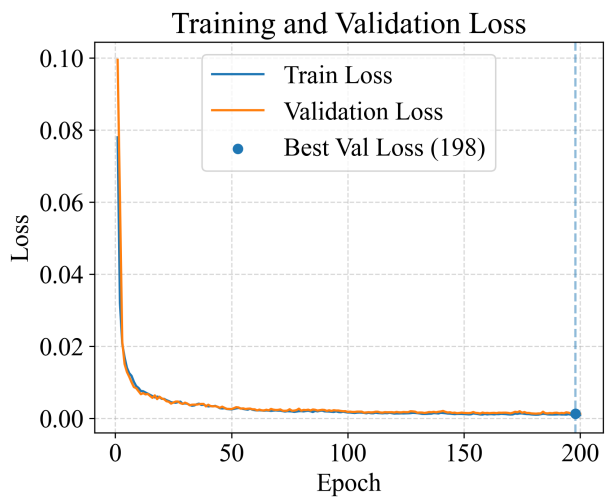


Fig. 2. Training loss for the DRCA used in generator-tripping cases.

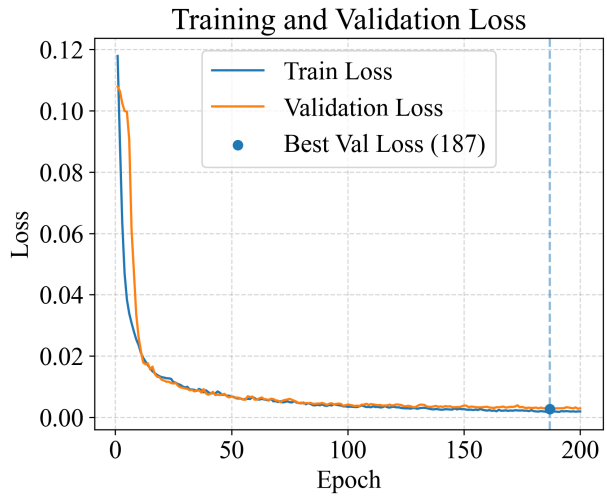


Fig. 3. Training loss for the DRCA used in load-shedding cases.